Balanced EF1 for Four Agents in Two Fixed Groups

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Abstract

We study indivisible goods allocated to two fixed nonempty groups whose members may value their common bundle differently. We require both envy-freeness up to one good (EF1) for every member and balancedness: the two bundles have nearly equal sizes. For arbitrary normalized monotone valuations, we prove that these requirements are compatible whenever there are at most four agents. This extends the balanced guarantee of Gölz and Yaghoubizade from additive valuations and adds balancedness to the monotone doubly-EF1 result of Bu et al.

The characterization of Kyropoulou, Suksompong, and Voudouris reduces the allocation theorem to a chromatic lower bound. We prove that the generalized Kneser graph $\mathrm{KG}(2q, q, 2)$ has chromatic number at least five for every $q \geq 2$, improving the previous lower bound of four. The proof straightens loops and spheres in the graph's neighborhood complex by coordinating cardinality-preserving exchanges, then contracts the resulting maps by a fixed sequence of shifts. Known six-agent counterexamples leave five agents as the only unresolved threshold; the five-agent guarantee is equivalent to the conjecture that these graphs are six-chromatic. Exact homology computations show that the neighborhood complexes for $q = 3, 4$ are not 3-connected, so the connectivity bound used here cannot supply the sixth color in those cases.

Keywords

Fair division, Fixed groups, Balanced allocations, Monotone valuations, EF1, Generalized Kneser graphs, Topological combinatorics

1 Introduction

Fair division among fixed groups differs from individual allocation because one bundle must satisfy several members with potentially different preferences. In the individual model, EF1 always exists under arbitrary monotone valuations [12]. The same is true when the allocator may choose the groups together with the bundles [6]. We study the harder *fixed-group* model: the two nonempty groups are prescribed, and fairness must hold for every member rather than for an aggregate group preference [11, 14].

An allocation is *envy-free up to one good* (EF1) if each agent's envy can be eliminated by hypothetically deleting at most one good from the other group's bundle. We additionally require *balancedness*: the bundle sizes differ by at most one. Balancedness is the group analogue of a cardinality constraint [2, 21]; it imposes nearly equal item quotas even when the group sizes differ. Values are normalized and monotone but need not be additive, so they may reflect interactions among goods.

Our first result shows that balance and every member's EF1 requirement are compatible for at most four agents, regardless of the number of goods.

Theorem 1.1 (Balanced EF1 for at most four agents). *For two nonempty fixed groups G_1, G_2 with $|G_1| + |G_2| \leq 4$ and arbitrary normalized, nonnegative monotone valuations, there exists a balanced EF1 allocation of any finite set of goods.*

For four agents, this covers both group-size splits, $(2, 2)$ and $(1, 3)$. Without the balance constraint, Bu et al.'s argument already gives EF1 for at most four monotone agents in every split [4, Theorem 3.5]; the new fair-division guarantee here is therefore precisely balancedness. It also makes the threshold split-independent. In the unbalanced model, KSV show for six binary agents that the $(5, 1)$ split always admits EF1, whereas counterexamples exist for $(4, 2)$ and $(3, 3)$. With balance, four agents always admit EF1 by our theorem, while KSV's four-good construction rules it out for six agents in every nonempty split. The five-agent case is equivalent to a 2017 chromatic-number conjecture, as KSV already observed; see Section 3.

1.1 The chromatic threshold

For a positive integer s, write $[s] = \{1, \ldots, s\}$, and let $\binom{U}{k}$ denote the k-subsets of U. Fix $q \geq 2$ and identify $2q$ goods with $[2q]$. Every balanced allocation is (A, A^c) for a unique $A \in \mathcal{V}_q = \binom{[2q]}{q}$, where A is assigned to the first group. Let $H_q = \mathrm{KG}(2q, q, 2)$ be the graph on $\mathcal{V}_q$ in which distinct A, B are adjacent, written $A \sim B$, when $|A \cap B| \leq 1$.

Kyropoulou, Suksompong, and Voudouris (KSV) [11] proved that, for any fixed monotone agent, the balanced allocations she rejects form an independent set of H_q. Consequently, if every balanced allocation were rejected, assigning each allocation one rejecting agent would properly color H_q. Thus $\chi(H_q) \geq 5$ prevents four agents from rejecting all balanced allocations. The previous lower bound $\chi(H_q) \geq 4$ [5] yields only the three-agent guarantee. Our main combinatorial result supplies the fifth color.

Theorem 1.2 (Chromatic lower bound). *For every integer $q \geq 2$,*

$$\chi\big(\mathrm{KG}(2q, q, 2)\big) \geq 5.$$

Proposition 3.2 and Theorem 1.2 prove Theorem 1.1 for an even number of goods; the standard dummy-good reduction handles the odd case in Section 6. The same proposition also shows that a proper coloring yields monotone valuations blocking every balanced allocation.

1.2 Relation to prior work

Doubly EF1 and allocator preferences. Bu, Li, Liu, Song, and Tao [3, 4] study two recipient agents with valuations v_1, v_2 and a single

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allocator with recipient-specific valuations u_1, u_2. An allocation (A_1, A_2) is *doubly EF1* if EF1 holds for v_1, u_1 with own bundle A_1 and for v_2, u_2 with own bundle A_2. The four removal witnesses may differ. Thus the coloring argument involves four valuation requirements. The WINE 2023 version pursued a balanced result using the conjectural equality $\chi(H_q) = 6$. The corrected full version instead establishes doubly EF1 for arbitrary monotone valuations without requiring balancedness [4, Theorems 3.1 and 3.5]. For $m \geq 3$ goods, it uses the graph $\Gamma(m)$ on all subsets of the goods, where distinct bundles A, B are adjacent when $|A \cap B| \leq 1$ and $|A \cup B| \geq m - 1$. They prove $\chi(\Gamma(m)) \geq 5$ using Lovász's neighborhood-complex theorem.

For $m = 2q$, the subgraph of $\Gamma(2q)$ induced by the q-subsets is exactly H_q, since $|A \cup B| = 2q - |A \cap B|$ on this layer. The lower bound for $\Gamma(2q)$ does not imply the same bound for its induced subgraph. Their correction explicitly identifies $\chi(H_q) \geq 5$ as the missing bound needed for balanced doubly EF1 [4, Section 3.2]. At a high level, both proofs follow the same deformation architecture. Bu et al. first remove their "terrible triangles," then their "bad edges," and finally contract to the cone-like vertex $\emptyset$. The balanced layer contains no such vertex. After straightening maps into the Johnson clique complex, we instead contract by a fixed sequence of cardinality-preserving shifts, using contiguity in the larger neighborhood complex. Moreover, straightening a sphere requires coordinated midpoints along shared edges because the Johnson subcomplex is a clique complex rather than a full skeleton. These are the new steps that establish $\chi(H_q) \geq 5$ and hence balanced doubly EF1 (Corollary 6.1).

Additive valuations. An *additive* valuation assigns each bundle the sum of its individual-good values. Gölz and Yaghoubizade establish balanced EF1 for four additive agents in both $(2, 2)$ and $(1, 3)$ groups [7, Section 3.1]. Their polynomial-time algorithm first constructs a *fractional allocation*, allowing goods to be divided between groups, and then rounds it to whole goods. Theorem 1.1 extends their existence guarantee to arbitrary monotone valuations, but does not extend their algorithm. The example in Section 2 illustrates why individual-good values no longer suffice to test EF1.

Other constraints and group guarantees. Cardinality constraints for individual recipients are studied by Biswas and Barman and surveyed by Suksompong [2, 21]. Here, by contrast, each bundle must satisfy an entire group. Other work studies *democratic fairness*, requiring a prescribed fraction of each group's members to be satisfied [20], random additive instances [14], or guarantees allowing more than one good to be removed from an envied bundle [15]. Our requirement is EF1 for every member of the two fixed groups, in every monotone instance.

The graph-theoretic bounds. Jafari and Alipour [1, Theorem 2.2] give $\chi(H_q) \leq 6$ for every $q \geq 2$ and conjecture equality for $q \geq 3$. Equality is immediate for $q = 2$, since H_2 is complete on six vertices. They also record $\chi(H_3) = 6$ and compute $\chi(H_4) = 6$ [1, Sections 3 and 5.2]. For this family, Daneshpajouh's general lower bound gives only $\chi(H_q) \geq 4$ [5, Theorem 1]. Theorem 1.2 raises this to five, leaving the distinction between five and six colors for $q \geq 5$.

1.3 Proof idea and organization

We associate with H_q its neighborhood complex $\mathcal{N}_q$: a family of balanced bundles spans a simplex when all its members have a common neighbor in H_q. Lovász's theorem yields $\chi(H_q) \geq 5$ once $|\mathcal{N}_q|$ is 2-connected, meaning that it is path-connected and every map from S^1 or S^2 is null-homotopic.

A family spans a simplex of $\mathcal{N}_q$ exactly when all its bundles lie within one exchange of a common reference bundle. This supports two steps. First, we replace every distance-two image edge by a path of single exchanges, choosing the replacements consistently along shared triangle boundaries. Second, a fixed sequence of cardinality-preserving shifts moves every resulting bundle toward one reference bundle while the homotopy remains in $\mathcal{N}_q$. Sections 4–5 formalize these steps.

Exact computations later show that the stronger 3-connectivity certificate fails for $q = 3, 4$, even though both graphs are six-chromatic. This limits the specific connectivity bound used here, not stronger topological lower bounds.

2 Model and Notation

Let M be a finite set of goods, with $m = |M|$. A *bundle* is a subset of M. The agents form two prescribed nonempty groups G_1, G_2, with sizes n_1, n_2 and total size $N = n_1 + n_2$. Each agent i has a valuation $v_i : 2^M \to \mathbb{R}_{\geq 0}$, where 2^M denotes the family of subsets of M and $\mathbb{R}_{\geq 0}$ the nonnegative reals. We assume normalization, $v_i(\emptyset) = 0$, and monotonicity, $v_i(S) \leq v_i(T)$ whenever $S \subseteq T \subseteq M$. When additivity is assumed, it means $v_i(S) = \sum_{x \in S} v_i(\{x\})$.

An allocation is a partition (A_1, A_2) of M, with A_ℓ assigned to group G_ℓ for $\ell \in \{1, 2\}$. It is *balanced* if $\big||A_1| - |A_2|\big| \leq 1$, and *EF1* if, for every $\ell \in \{1, 2\}$ and $i \in G_\ell$, there exists $D \subseteq A_{3-\ell}$ with $|D| \leq 1$ such that

$$v_i(A_\ell) \geq v_i(A_{3-\ell} \setminus D). \tag{1}$$

The *EF1 witness* D may be empty and may differ between agents. An agent *rejects* an allocation if no such witness exists.

Write $A^c = M \setminus A$. When $m = 2q \geq 4$, label the goods by $[2q]$. A balanced allocation then has the form (A, A^c) with $A \in \mathcal{V}_q$, using the notation of Section 1.1. Our graph H_q, whose adjacency condition is $|A \cap B| \leq 1$, is denoted $\mathrm{KG}^2(2q, q, 1)$ by Daneshpajouh [5] and $\mathcal{K}(2q, q, 1)$ by Bu et al. [4].

Why full bundle values matter. The fractional relaxation used for additive valuations has no direct meaning for an arbitrary set function, and singleton values need not determine bundle values. The following four-good example also shows why greedy picking and a singleton-sum EF1 test can fail.

Let $M = \{a, b, c, d\}$, $G_1 = \{1, 2\}$, and $G_2 = \{3, 4\}$. Agents $1, 2, 3$ are additive: they value a, c, b, respectively, at 4 and all other goods at zero. Set $(w_a, w_b, w_c, w_d) = (2, 1, 2, 1)$. For a condition E, let $\mathbf{1}_E$ equal 1 when E holds and 0 otherwise. Agent 4 assigns each bundle $S \subseteq M$ the value

$$v_4(S) = \max_{x \in S} w_x + 3\,\mathbf{1}_{\{a,c\} \subseteq S},$$

where the maximum over the empty set is zero. This valuation is normalized and monotone, but not additive.

Suppose agents pick once each in order $1, 3, 2, 4$, each maximizing the increase in the value of her group's bundle according to her

own valuation. They choose a, b, c, d, respectively, producing the balanced allocation $(\{a, c\}, \{b, d\})$. Agent 4 receives value 1, but removing either good from the other bundle leaves value 2:

$$v_4(\{b, d\}) = 1 < 2 = v_4(\{a\}) = v_4(\{c\}).$$

Thus this allocation is not EF1.

Replacing v_4 by the sum of its singleton values would incorrectly certify EF1: the own bundle and either one-good remainder of the other bundle would all have value 2. Thus the additive relaxation and picking rules cannot be transferred by simply substituting singleton values; the monotone problem requires the actual values of whole bundles.

3 The Chromatic Characterization

If $A \sim B$ in H_q, then $A \cup B$ misses at most one good. The resulting containment relations let monotonicity show that a fixed agent cannot reject both corresponding allocations. This is the key lemma behind KSV's characterization [11, Theorem 4.4]; it is also the balanced specialization of Bu et al.'s Proposition 3.2 [4].

Lemma 3.1 (EF1 violations are independent). *For $|M| = 2q \geq 4$, a group G_ℓ, and a fixed monotone agent $i \in G_\ell$, the vertices $A \in \binom{M}{q}$ for which (A, A^c) violates agent i's EF1 condition form an independent set of H_q.*

Proof. Suppose $A \sim B$. Set $D = M \setminus (A \cup B)$. Since $|A \cup B| \geq 2q-1$, we have $|D| \leq 1$ and $D \subseteq A^c \cap B^c$. If an agent in G_1, with valuation v, rejected both allocations, then

$$v(A) < v(A^c \setminus D) \leq v(B) < v(B^c \setminus D) \leq v(A).$$

The strict inequalities express failure of (1); the weak ones use $A^c \setminus D \subseteq B$ and $B^c \setminus D \subseteq A$. This is a contradiction. For an agent in G_2, apply the same argument to A^c, B^c, since $|A^c \cap B^c| = |A \cap B| \leq 1$. □

If every balanced allocation is rejected, labeling it by a rejecting agent gives a proper coloring. Conversely, each color class can be realized by a monotone agent who rejects all allocations in that class. The following is KSV's Theorem 4.4 and the tightness construction immediately following it, reformulated as an equivalence; we include the short proof for completeness.

Proposition 3.2 (KSV threshold characterization, reformulated). *Fix $q \geq 2$ and an integer $d \geq 2$. The following statements are equivalent:*

1. *$\chi(H_q) \geq d$.*
2. *For every $n_1, n_2 \geq 1$ with $n_1 + n_2 < d$, every normalized, nonnegative monotone instance on $2q$ goods with fixed group sizes (n_1, n_2) admits a balanced EF1 allocation.*

Proof. For the forward direction, an instance with no balanced EF1 allocation would give a proper coloring with $n_1 + n_2 < d$ colors by Lemma 3.1.

Conversely, suppose $\chi(H_q) < d$, and take an optimal proper coloring with $t = \chi(H_q) < d$ nonempty classes. Since H_q has an edge, $t \geq 2$. Create one agent per class, assigning at least one to each group. For a class C assigned to G_2, define

$$v_C(S) = \begin{cases} 1, & |S \cap B| \geq q - 1 \text{ for some } B \in C, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

This valuation is monotone and normalized because $q \geq 2$. If G_1 receives $A \in C$, then $|A \cap B| \geq 2$ for every $B \in C$: this is immediate for $B = A$, and for $B \neq A$ it follows because a color class is independent. Hence $|A^c \cap B| \leq q - 2$, so the agent values her own bundle A^c at zero. But A and every $A \setminus \{x\}$, $x \in A$, have value one. She therefore rejects this allocation.

For a class assigned to G_1, use (2) with $\{B^c : B \in C\}$ in place of C. The identity $|A^c \cap B^c| = |A \cap B|$ for q-sets gives the same conclusion. Every balanced allocation belongs to a class and is rejected by its agent. Hence the t-agent instance has no balanced EF1 allocation. □

The blocking valuations are 0–1 set functions, not necessarily additive. The characterization therefore does not identify the threshold for additive instances.

The six-agent limit. KSV's tightness construction gives counterexamples on four goods [11, following Theorem 4.4]. Here H_2 is complete on six vertices. Index six agents by the two-element subsets A of $M = [4]$ and distribute them in any prescribed nonempty split. An agent A in G_2 has value one exactly for bundles meeting A; in G_1, replace A by A^c. At allocation (A, A^c), the indexed agent values her own bundle at zero and the other at one after either deletion. Thus every balanced allocation is rejected.

Only five agents remain unresolved. Proposition 3.2 and the upper bound $\chi(H_q) \leq 6$ show that a universal five-agent guarantee is equivalent to the Jafari–Alipour conjecture: $\chi(H_q) = 6$ for every $q \geq 2$. KSV already recorded this implication [11, following Theorem 4.4]; the $q = 2$ case is immediate.

4 Topology and the Geometry of Balanced Bundles

We prove Theorem 1.2 by translating almost-disjointness in H_q into one-exchange geometry in its neighborhood complex. We introduce only the required topology; standard background appears in Matoušek [16, Chapters 1 and 4].

4.1 Simplices and deformations

A finite *simplicial complex* K on a vertex set V is a family of subsets closed under taking subsets and containing every singleton. Its members are *simplices*; their subsets are *faces*. For $U \subseteq V$, the *full simplex* $\Delta(U)$ consists of all subsets of U. The *geometric realization* $|K|$ is obtained by placing the vertices as basis vectors in $\mathbb{R}^V$ and filling the convex hull of each simplex.

A *simplicial map* $f : K \to K'$ sends vertices to vertices and simplices to simplices; its realization $|f| : |K| \to |K'|$ is linear on each simplex. Continuous maps are *homotopic* if one deforms continuously into the other, and *null-homotopic* if homotopic to a constant. A space is *k-connected* if it is path-connected and every map from S^r is null-homotopic for $1 \leq r \leq k$.

A *triangulation* expresses a space as a simplicial realization, and a *subdivision* refines it without changing the space. Simplicial approximation lets us replace a continuous map from a circle or sphere by a homotopic simplicial map after subdividing its domain. Every edge of a triangulated S^2 has exactly two incident faces, regardless of overlaps in their images.

Our homotopies use straight-line interpolation. Simplicial maps $f, g : K \to K'$ are *contiguous* if $f(\sigma) \cup g(\sigma)$ is a simplex of K' for every $\sigma \in K$; interpolation inside that target simplex gives a homotopy. The same interpolation applies after passing to a common subdivision when both maps send each original face into one target simplex.

4.2 The neighborhood complex and Lovász's bound

Write $V(H)$ for the vertex set of a graph H. A vertex is *isolated* if it has no neighbor. For a graph with no isolated vertices, its *neighborhood complex* $\mathcal{N}(H)$ consists of the families in $V(H)$ having a common neighbor, together with the empty set. Every vertex of H_q is adjacent to its complement, so this defines $\mathcal{N}_q = \mathcal{N}(H_q)$ on all of $\mathcal{V}_q$.

Theorem 4.1 (Lovász [13]). *For an integer $k \geq 0$, if $|\mathcal{N}(H)|$ is k-connected, then $\chi(H) \geq k + 3$.*

We prove that $|\mathcal{N}_q|$ is 2-connected; the theorem will then give $\chi(H_q) \geq 5$.

4.3 One exchange at a time: Johnson geometry

The *Johnson graph* $J(2q, q)$ has vertex set $\mathcal{V}_q$ and joins bundles differing by one exchange. Its graph distance, the least number of edges in a joining path, is $d_J(A, B) = q - |A \cap B|$. The closed radius-one ball centered at $Z \in \mathcal{V}_q$ is

$$B_1(Z) = \{A \in \mathcal{V}_q : d_J(A, Z) \leq 1\}.$$

Lemma 4.2 (Common neighbors are Johnson balls). *A set $\sigma \subseteq \mathcal{V}_q$ is a simplex of $\mathcal{N}_q$ if and only if $\sigma \subseteq B_1(Z)$ for some $Z \in \mathcal{V}_q$. Consequently, the full simplex $\Delta(B_1(Z))$ is a subcomplex of $\mathcal{N}_q$.*

Proof. For $Y \in \mathcal{V}_q$, put $Z = Y^c$. For $A \in \mathcal{V}_q$, the condition $A \sim Y$ is equivalent to $|A \cap Z| \geq q - 1$, or $d_J(A, Z) \leq 1$. This excludes $A = Y$ automatically, since $d_J(Y, Y^c) = q \geq 2$. Thus Y is a common neighbor of σ exactly when $\sigma \subseteq B_1(Z)$. □

The three edge relations play different roles: H_q joins almost disjoint bundles, Johnson edges represent single exchanges, and edges of $\mathcal{N}_q$ may join bundles two exchanges apart. A whole Johnson ball fills a simplex of $\mathcal{N}_q$.

For example, take $q = 3$ and $Z = \{1, 2, 3\}$. The ball $B_1(Z)$ contains Z and the nine bundles obtained by replacing one of $1, 2, 3$ by one of $4, 5, 6$. All ten share the neighbor $Z^c = \{4, 5, 6\}$ and fill a simplex of $\mathcal{N}_3$. The bundles $A = \{1, 2, 4\}$ and $B = \{1, 3, 5\}$ lie in this ball but have distance two. The filled triangle on A, Z, B deforms the edge AB into the one-exchange path $A - Z - B$. For a sphere, such repairs must agree on shared boundaries.

A *clique* is a family of pairwise adjacent vertices. The *clique complex* consists of a graph's cliques, singletons, and the empty set. Let L_q be the clique complex of $J(2q, q)$. Its simplices have pairwise Johnson distances at most one, so each nonempty simplex lies in the ball centered at any of its vertices. Hence $L_q \subseteq \mathcal{N}_q$.

We first move maps of loops and spheres into $|L_q|$, then contract the inclusion $|L_q| \hookrightarrow |\mathcal{N}_q|$. The contraction takes place in the larger complex $\mathcal{N}_q$; L_q itself need not be contractible.

5 The Chromatic Lower Bound

Every edge of $\mathcal{N}_q$ has Johnson distance at most two by Lemma 4.2. Call an image edge an *exchange edge* if its endpoints have distance at most one, allowing coincident images. We first replace distance-two edges by exchange paths, compatibly across shared faces (Proposition 5.2). We then contract the repaired map by a fixed exchange sequence (Proposition 5.6).

5.1 The four possible midpoints

For bundles at distance two, a *common midpoint* is a bundle at distance one from each.

Lemma 5.1 (Four common midpoints). *Suppose $d_J(A, B) = 2$. Write $A = X \cup \{a, b\}$ and $B = X \cup \{c, d\}$, where $|X| = q - 2$ and a, b, c, d are distinct and outside X. The vertices at distance at most one from both are exactly*

$$\begin{aligned} Q_{ac} &= X \cup \{a, c\}, \quad Q_{ad} = X \cup \{a, d\}, \\ Q_{bc} &= X \cup \{b, c\}, \quad Q_{bd} = X \cup \{b, d\}. \end{aligned} \tag{3}$$

They form a four-cycle in the Johnson graph. Each midpoint is at distance two from its opposite and at distance at most one from the other two.

Proof. A common midpoint differs from A by one exchange. To increase its intersection with B from $q - 2$ to at least $q - 1$, it must delete a or b and insert c or d, giving (3). Two displayed midpoints have distance one exactly when their two-element suffixes share one of a, b, c, d; otherwise they are opposite and have distance two. □

For a loop, any midpoint works. For a sphere, a shared edge needs one midpoint lying in both incident face balls. Repairing faces independently could give different paths along their common boundary. We therefore choose shared midpoints before filling the faces.

5.2 Straightening loops and spheres

Proposition 5.2 (Straightening). *For $r = 1, 2$ and every continuous map $f : S^r \to |\mathcal{N}_q|$, there is a continuous map $g : S^r \to |L_q|$ such that $f \simeq \iota \circ g$ in $|\mathcal{N}_q|$, where $\iota : |L_q| \hookrightarrow |\mathcal{N}_q|$ is the inclusion.*

Proof. After subdividing the domain as needed, simplicial approximation reduces the claim to the realization of a simplicial map $f : T \to \mathcal{N}_q$, where T triangulates S^r. Identify $|T|$ with S^r. For $r = 2$, every domain edge has exactly two incident triangular faces.

Loops. For an edge with image endpoints A, B at distance two, insert a vertex mapped to a common midpoint Q and replace AB by $A{-}Q{-}B$. Both new edges are exchange edges. Since $A, Q, B \in B_1(Q)$, linear interpolation in $|\Delta(B_1(Q))|$ deforms the old edge map to the new path while fixing endpoints. These homotopies agree at the original vertices, giving a loop in $|L_q|$.

Spheres: choose face centers. Let $r = 2$. For each triangular face $\tau = \{u, v, w\}$ of T, choose $Z_\tau \in \mathcal{V}_q$ with

$$f(\tau) \subseteq B_1(Z_\tau), \tag{4}$$

using Lemma 4.2. The entire continuous image $|f|(|\tau|)$ then lies in the convex simplex $|\Delta(B_1(Z_\tau))|$.

Choose shared midpoints. Let $e = \{u, v\}$ have image endpoints $A = f(u)$, $B = f(v)$ at distance two. Its two incident faces τ_1, τ_2 have centers that are common midpoints of A, B, by (4). These centers may coincide. Each excludes only its opposite among the four midpoints in Lemma 5.1, leaving at least two choices for a common midpoint Q_e satisfying

$$d_J(Q_e, Z_{\tau_i}) \le 1 \qquad (i = 1, 2). \tag{5}$$

Insert one vertex p_e on e and map it to Q_e, using this same vertex and image in both faces. Leave exchange edges unchanged. Choices on distinct edges are independent because the face centers are already fixed.

For example, let $q = 3$, $A = \{1, 2, 3\}$, and $B = \{1, 4, 5\}$. The centers $\{1, 2, 4\}$ and $\{1, 3, 5\}$ are opposite midpoints. Neither center works for the other face, but either remaining midpoint, $\{1, 2, 5\}$ or $\{1, 3, 4\}$, works for both.

Fill the faces. Each original face τ now has a boundary of three to six sides. All image vertices lie in $B_1(Z_\tau)$, by (4)–(5), and all image edges are exchange edges. Insert an interior vertex w_τ and join it to every boundary vertex; this is *coning the boundary* (Figure 1). Map w_τ to Z_τ. The endpoints U, V of each image boundary edge are within one exchange of each other and of Z_τ, so each new triangle maps into L_q.

The constructions agree on shared edges and define a simplicial map $g : T' \to L_q$ on a refinement T' of T. This is still a triangulated sphere: each subdivided original edge has one triangle on each side, and each new interior edge has two incident triangles within its original face.

Verify the homotopy. Identify $|T'|$ with $|T| = S^2$. Over each original face τ, both $|f|$ and $|g|$ take values in the same convex simplex $|\Delta(B_1(Z_\tau))|$. For $x \in S^2$ and $t \in [0, 1]$, set

$$F(x, t) = (1 - t)|f|(x) + t|g|(x).$$

The segment stays in that simplex. Since $|f|$ and $|g|$ are globally continuous in the common Euclidean realization, so is F, including across face boundaries. Its endpoints are $|f|$ and $\iota \circ |g|$, giving the required homotopy. □

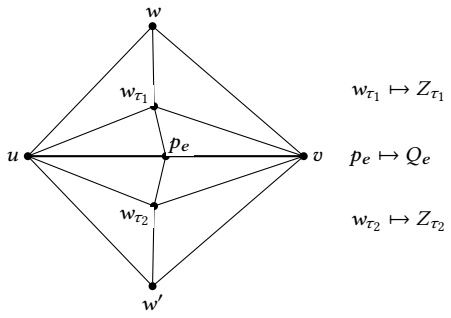


Figure 1: Two incident domain faces $\tau_1 = \{u, v, w\}$ and $\tau_2 = \{u, v, w'\}$. Their shared edge $e = \{u, v\}$ has one new vertex p_e, with the same image Q_e from both sides. Each interior vertex maps to its own face center. The midpoint condition puts Q_e within one exchange of both centers. The drawing shows the case in which the four outer edges already have one-exchange images.

Only the two faces incident to a domain edge constrain its midpoint; overlaps among image faces impose no additional conditions.

5.3 Cardinality-preserving shifts

For distinct $a, b \in [2q]$, define the *shift* on $A \in \mathcal{V}_q$ by

$$S_{a\to b}(A) = \begin{cases} (A \setminus \{a\}) \cup \{b\}, & a \in A,\ b \notin A, \\ A, & \text{otherwise.} \end{cases}$$

We need two properties: shifts preserve Johnson cliques, so they can be composed on L_q; and each simplex together with its shifted image lies in one simplex of $\mathcal{N}_q$, where interpolation gives a homotopy.

Lemma 5.3 (Shifts do not increase distance). *For all $A, B \in \mathcal{V}_q$,*

$$d_J(S_{a\to b}(A), S_{a\to b}(B)) \le d_J(A, B).$$

In particular, $S_{a\to b}$ is a simplicial map from L_q to itself.

Proof. Represent a set by its 0–1 membership vector. Johnson distance on q-sets is half the Hamming distance, which counts differing coordinates. On coordinates a, b, the shift sends 10 to 01 and fixes 00, 01, 11. The distances from 10 to those fixed patterns change from $1, 2, 1$ to $1, 0, 1$; all other pairwise distances are unchanged. Thus distance cannot increase, and pairwise distance at most one is preserved. □

For $W, U \subseteq [2q]$ with $|W| = q - 1$ and $|U| = q + 1$, define the *star* and *top*

$$\mathcal{S}(W) = \{A \in \mathcal{V}_q : W \subseteq A\}, \qquad \mathcal{T}(U) = \{A \in \mathcal{V}_q : A \subseteq U\}.$$

A star's bundles share $q - 1$ goods; a top's lie within $q + 1$ goods.

Lemma 5.4 (Stars and tops). *Every clique of $J(2q, q)$ is contained in a star or a top.*

Proof. The claim is immediate for at most two vertices. Otherwise choose $A = I \cup \{a\}$ and $B = I \cup \{b\}$ in the clique, with $|I| = q - 1$ and $a \ne b$. Another clique vertex C either contains I, or omits some $x \in I$. In the latter case, to meet both A and B in $q - 1$ elements, it

must contain $I \setminus \{x\}$ and both a, b. Hence $C = (I \cup \{a, b\}) \setminus \{x\}$, the top alternative.

A star vertex other than A, B is $I \cup \{y\}$ with $y \notin I \cup \{a, b\}$. Its distance to a top vertex other than A, B is two. Thus the clique cannot mix the two alternatives beyond A, B. □

Lemma 5.5 (Contiguity). *As maps $L_q \to \mathcal{N}_q$, the inclusion and $S_{a\to b}$ are contiguous: $\sigma \cup S_{a\to b}(\sigma) \in \mathcal{N}_q$ for every simplex σ of L_q.*

Proof. By Lemma 5.4, σ lies in a star or a top. First suppose $\sigma \subseteq \mathcal{S}(W)$. If $b \in W$, the shift fixes every vertex. If $a, b \notin W$, it only replaces $W \cup \{a\}$ by $W \cup \{b\}$, so the union remains in the star and hence in a common ball.

If $a \in W$ and $b \notin W$, take $Z = W \cup \{b\}$. Every original vertex lies within one exchange of Z. Each changed vertex has the form $(W \setminus \{a\}) \cup \{b, x\}$ and is also within one exchange of Z; the only unchanged vertex is Z. Thus the union lies in $B_1(Z)$ and is a simplex of $\mathcal{N}_q$.

For a top, take complements. Complementation preserves Johnson distance and maps $B_1(Z)$ to $B_1(Z^c)$, so it is a simplicial automorphism of $\mathcal{N}_q$. It sends tops to stars and satisfies $(S_{a\to b}(A))^c = S_{b\to a}(A^c)$. The star case therefore applies. □

Why the ambient space is needed. For $q = 3$, apply $S_{1\to 3}$ to the Johnson edge with endpoints $\{1, 2, 4\}$ and $\{1, 2, 5\}$. The images are $\{2, 3, 4\}$ and $\{2, 3, 5\}$. All four bundles lie in $B_1(\{1, 2, 3\})$, hence span a simplex of $\mathcal{N}_3$. But $\{1, 2, 5\}$ and $\{2, 3, 4\}$ have distance two, so the union is not a simplex of L_3. The homotopy needs the larger complex.

Proposition 5.6 (Contracting the inclusion). *The inclusion $|L_q| \hookrightarrow |\mathcal{N}_q|$ is null-homotopic.*

Proof. Fix $R = \{1, \dots, q\}$. For each $e \in [2q] \setminus R$, apply the block $S_{e\to 1}, \dots, S_{e\to q}$. Concatenate the blocks in a fixed order, using this same sequence of q^2 shifts on every vertex; denote it by $T_1, \dots, T_{q^2}$. Let F_0 be the identity on L_q, define $F_{j+1} = T_{j+1} \circ F_j$, and set $F_* = F_{q^2}$. Each F_j maps L_q simplicially to itself by Lemma 5.3.

Lemma 5.5 applied to $F_j(\sigma)$ gives

$$F_j(\sigma) \cup F_{j+1}(\sigma) \in \mathcal{N}_q \qquad (\sigma \in L_q).$$

Thus successive maps are contiguous into $\mathcal{N}_q$, and their homotopies concatenate from the inclusion to F_*.

To see that F_* is constant, consider the block for e. If e is present, some element of R is absent. Until a missing element is reached, the block leaves the bundle unchanged; that exchange removes e. Later shifts never reinsert outside elements. After all blocks, the bundle has q elements, all in R, so equals R. Therefore F_* is constant on vertices and on the realization. □

Thus one fixed shift sequence contracts the inclusion, and every vertex image throughout the simplicial sequence remains a q-subset.

5.4 Completing the connectivity argument

Proposition 5.7 (Two-connectivity). *For every $q \geq 2$, $|\mathcal{N}_q|$ is 2-connected.*

Proof. The Johnson graph is connected: repeatedly exchange an unwanted element for a missing one to transform any q-set into any other. Since $L_q \subseteq \mathcal{N}_q$ and both have the same vertices, these paths connect all vertices of $\mathcal{N}_q$. Every other point joins a vertex within its simplex, proving path-connectedness.

Let $\iota : |L_q| \hookrightarrow |\mathcal{N}_q|$ be the inclusion. For $r = 1, 2$, Proposition 5.2 deforms any continuous $f : S^r \to |\mathcal{N}_q|$ to $\iota \circ g$ for some continuous $g : S^r \to |L_q|$. Proposition 5.6 contracts ι, so $\iota \circ g$ and hence f are null-homotopic. □

Proof of Theorem 1.2. Proposition 5.7 and Theorem 4.1, with $k = 2$, give $\chi(H_q) \geq 5$. □

6 Fair-Division Consequences

The chromatic characterization now converts the five-color bound into the fair-division theorem; a dummy good handles odd cardinalities.

Proof of Theorem 1.1. For $|M| \leq 2$, every balanced allocation is EF1 because each bundle has at most one good. For $|M| = 2q \geq 4$, Theorem 1.2 and Proposition 3.2, with $d = 5$, give balanced EF1 for $n_1 + n_2 \leq 4$.

For odd $|M| \geq 3$, use KSV's dummy-good reduction [11, Theorem 4.4]: add a good δ and set $v_i'(S) = v_i(S \setminus \{\delta\})$. These valuations remain normalized and monotone. Apply the even case and remove δ. Bundle sizes then differ by one. If D is an EF1 witness in the augmented instance, then $D' = D \setminus \{\delta\}$ is a witness after deleting the dummy good, so EF1 is preserved. □

6.1 Balanced doubly EF1

Return to the two recipient valuations v_1, v_2 and the allocator's recipient-specific valuations u_1, u_2 from Section 1.2 [4].

Corollary 6.1 (Balanced doubly EF1). *For arbitrary normalized, nonnegative monotone valuations v_1, v_2, u_1, u_2, a balanced doubly EF1 allocation exists.*

Proof. Assign v_1, u_1 to G_1 and v_2, u_2 to G_2, and apply Theorem 1.1. □

7 Algorithmic Perspective

The existence proof yields an implicit coloring but not, in general, an efficient way to locate a monochromatic edge. Let $N = n_1 + n_2 \leq 4$, with agents labeled by $[N]$, and suppose a *value oracle* returns $v_i(S)$ on a query (i, S). A proposed balanced allocation and its EF1 witnesses can be verified with two value queries per agent.

Proposition 7.1 (Oracle search reduction). *For $m = 2q \geq 4$, balanced-EF1 search reduces to finding a monochromatic edge in an implicit labeling $c : \mathcal{V}_q \to [N]$ of H_q. Each label evaluation takes at most $N(q + 1)$ value queries. From any such edge, a balanced EF1 allocation and its witnesses require no further queries if both endpoint evaluations are retained, and at most $2N(q + 1)$ queries otherwise.*

Proof. Let $c(A)$ be the first agent rejecting (A, A^c), or 1 if none does. To evaluate it, query each agent's own-bundle value and all q single-good deletions from the other bundle. These $q + 1$ queries decide EF1 and find a witness when one exists. Single deletions suffice: if the empty witness works, monotonicity makes any deletion work. Retain every agent's test outcome and witness with the label.

Since $N \leq 4 < \chi(H_q)$, there is an edge AB with $c(A) = c(B)$. If neither allocation were EF1 for all agents, their common label

would identify an agent rejecting both, contrary to Lemma 3.1. The retained outcomes identify a balanced EF1 endpoint and all its witnesses. Without those records, testing both endpoints costs at most $2N(q+1)$ queries. □

Odd m uses the dummy extension from Section 6.

Proposition 7.2 (Two-agent search). *For $N = 2$ and $m = 2q$, a balanced EF1 allocation and witnesses can be found with at most $2(q+1)(2q-1) = O(q^2)$ value queries.*

Proof. Fix $A_0 \in \mathcal{V}_q$ and a Johnson geodesic $A_0, A_1, \ldots, A_q = A_0^c$. The sequence

$$A_0^c, A_1, A_1^c, A_2, \ldots, A_{q-1}, A_{q-1}^c, A_0^c$$

is a closed walk of odd length $2q-1$ in H_q: complement pairs are disjoint, and $|A_i^c \cap A_{i+1}| = 1$. Every two-labeling of an odd closed walk has a monochromatic edge. Evaluate the labeling from Proposition 7.1 at the $2q-1$ vertices. Each evaluation uses at most $2(q+1)$ queries, and retained outcomes identify an EF1 endpoint and its witnesses without further queries. □

For three or four agents, finding a monochromatic edge with polynomially many label evaluations remains open. When $q \geq 4$, H_q is triangle-free: three pairwise intersections of size at most one would force their union to have size at least $3q-3 > 2q$. Thus a constant-size clique cannot provide the certificate used above.

A finite-input formulation. TFNP consists of total search problems with polynomial-size solutions and polynomial-time verification [18]. An oracle alone is not a finite encoding. For a concrete encoding, use an arithmetic circuit whose input bit x_g records whether good g belongs to the bundle. Gates are +, min, and max, with binary-encoded nonnegative integer constants. This syntax guarantees monotonicity; normalization is checked by evaluation at the empty bundle. Circuit values have polynomial bit length. A distinguished invalid-input output handles malformed circuits or circuits with nonzero empty-bundle value. On valid inputs, a solution is a balanced EF1 allocation with its witnesses, giving a TFNP problem.

PPA (Polynomial Parity Argument) is the TFNP subclass based on finding another odd-degree vertex in an implicitly represented undirected graph, given one [19]. Haviv proved the analogous Schrijver search problem PPA-complete [9]; restricting a Kneser coloring to the Schrijver graph places ordinary Kneser search in PPA. No hardness result is known for ordinary Kneser search [10]. For the intersection-at-most-one layer here, we are not aware of a Tucker–Fan-type combinatorial certificate; in particular, PPA membership remains open.

8 Limits of the Connectivity Certificate

Lovász's theorem would give six colors if $|\mathcal{N}_q|$ were 3-connected. Exact homology shows that this sufficient condition fails for the two smallest nontrivial six-chromatic cases.

Why the surface argument does not iterate. The straightening proof exploits a specifically two-dimensional incidence fact. An edge of a triangulated surface has exactly two incident faces. If its image endpoints are at Johnson distance two, they have four common midpoints, while the center chosen for each incident face excludes at most one midpoint. A shared choice therefore remains. In a triangulated three-manifold, by contrast, the link of an edge is a cycle and the edge can meet four or more tetrahedra. Four image centers can already exclude all four midpoints. Thus independent local choices need not agree; a higher-dimensional extension would require a global coordination argument or a different target subcomplex. The next computation shows that, for $q = 3, 4$, this is not merely a defect of our proof: the neighborhood complex itself is not 3-connected.

Over $\mathbb{F}_2$, let f_j be the number of j-simplices of $\mathcal{N}_q$ and let r_j be the rank of the boundary map $\partial_j : C_j \to C_{j-1}$. The jth Betti number is

$$b_j = \dim H_j(\mathcal{N}_q; \mathbb{F}_2) = f_j - r_j - r_{j+1}.$$

Computing b_3 therefore requires simplex counts and boundary ranks through dimension four.

Proposition 8.1 (Exact finite computation). *For $q = 3, 4$, the neighborhood complex $|\mathcal{N}_q|$ is not 3-connected. Exact computation over $\mathbb{F}_2$ gives*

q	f_3	r_3	r_4	b_3
3	2850	799	2040	11
4	129500	21209	108052	239

and $b_0 = 1$, $b_1 = b_2 = 0$ in both cases.

Proof (by computation). List the q-subsets of $[2q]$. For each center Z, generate all subsets of $B_1(Z)$ of size at most five and remove duplicates. By Lemma 4.2, this exhaustively lists the simplices through dimension four. The resulting face vectors are

q	f_0	f_1	f_2	f_3	f_4
3	20	180	960	2850	4500
4	70	1820	22960	129500	402472.

Exact sparse Gaussian elimination on the boundary matrices gives

q	r_1	r_2	r_3	r_4
3	19	161	799	2040
4	69	1751	21209	108052.

Two independent implementations agree on every displayed count and rank. One generates simplices from the radius-one balls and reduces columns in several orders; the other generates them from common-neighbor intersections and reduces transposed matrices with a different pivot convention. The identities forced by Proposition 5.7,

$$r_1 = f_0 - 1, \qquad r_2 = f_1 - r_1, \qquad r_3 = f_2 - r_2,$$

provide independent checks on all ranks through dimension three. Finally $b_3 = f_3 - r_3 - r_4$ is positive. A 3-connected space has zero third homology, by Hurewicz and universal coefficients [8], so neither complex is 3-connected. □

The next computational scale. For $q = 5$, the same enumeration is still feasible and gives

$$(f_0, f_1, f_2, f_3, f_4) = (252, 15750, 361200, 3156300, 15807960).$$

The first two exact ranks are $r_1 = 251$ and $r_2 = 15499$, confirming $b_0 = 1$ and $b_1 = 0$. Proposition 5.7 then forces $r_3 = f_2 - r_2 = 345701$. Determining b_3 still requires the much larger rank r_4, which we do not report. These partial data show that simplex generation

is practical at the next case, but they carry no conclusion about 3-connectivity.

Because $\chi(H_3) = \chi(H_4) = 6$ [1], the positive third homology rules out only the 3-connectivity certificate in Theorem 4.1. Indeed, the Matoušek–Ziegler hierarchy contains the chain

$$\chi(H) \geq \operatorname{ind}(B(H)) + 2 \geq \operatorname{coind}(B(H)) + 2 \geq \operatorname{conn}(N(H)) + 3,$$

where $B(H)$ is a box complex [17]. Consequently, failure of the last, weakest bound does not rule out a six-color lower bound from the $\mathbb{Z}_2$ index, coindex, or Stiefel–Whitney height. Nor does the computation address 3-connectivity for $q \geq 5$.

9 Conclusion and Further Questions

We proved $\chi(H_q) \geq 5$ for every $q \geq 2$ and thereby established balanced EF1 for at most four agents in two fixed groups under arbitrary monotone valuations. Together with the known six-agent counterexamples, this leaves exactly the five-agent case, equivalent to the six-color conjecture.

For six colors, 3-connectivity for every $q \geq 5$ remains one possible route, while the $q = 3, 4$ computations point toward stronger equivariant invariants or a different auxiliary complex for a uniform topological proof. The local obstruction above explains why the surface repair does not extend face by face; it does not exclude a coordinated construction.

Algorithmically, Proposition 7.2 settles the two-agent case. For three or four agents, can the monochromatic edge in Proposition 7.1 be found with polynomially many value queries? A positive answer would give a polynomial-query allocation procedure; polynomial running time would additionally require polynomial computation between queries.

This search question is distinct from the five-agent existence question. For three and four agents, existence is already established, but evaluating the implicit labeling on all $\binom{2q}{q}$ balanced bundles is exponential in q. The odd walk in Proposition 7.2 replaces that enumeration by a linear-size certificate for two labels. A comparable global certificate for three or four labels is unknown; triangle-freeness for $q \geq 4$ explains why the most immediate constant-clique analogue cannot work. Establishing a Tucker–Fan-type parity formulation would also place the finite-circuit version in PPA.

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